Embedding Migration Plan — Ollama to sentence-transformers

ISD Document Intelligence V5 | Objective: Eliminate HTTP bottleneck for bulk indexing | Date: March 2026

1. Why This Change?

Problem	Impact	Solution
Ollama embed API = one HTTP request per chunk	~200–500ms network overhead per call	sentence-transformers loads model directly on GPU
No native batching in Ollama /api/embeddings	GPU sits mostly idle between calls	Batch 256 chunks in one GPU call
Read timeouts on large files	Files fail and need manual retry	Direct GPU = no HTTP timeout

> **Same model:** mxbai-embed-large in Ollama is the same as `mixedbread-ai/mxbai-embed-large-v1` on HuggingFace. Both produce 1024-dimension vectors.

2. What Changes

Component	Before	After
Embedding (Indexing)	Ollama /api/embed (HTTP per chunk)	**sentence-transformers direct GPU (batch 256)**
Embedding (Query)	Ollama /api/embed (single call)	**sentence-transformers direct GPU (single call)**
LLM (Q&A, Entity extraction)	Ollama gemma3:12b	No change — still Ollama gemma3:12b
Vector Store	ChromaDB	No change — still ChromaDB
MySQL	All 10 tables	No change

3. Toggle Mechanism

> A single flag in `.env` controls which embedding method is used:
>
> `USE_OLLAMA_EMBEDDINGS=true` → Uses Ollama (current behavior)
> `USE_OLLAMA_EMBEDDINGS=false` → Uses sentence-transformers (direct GPU)

> This is fully reversible at any time.

4. Files Modified (3 files only)

File	Change
config.py	Add `USE_OLLAMA_EMBEDDINGS` flag (1 line)
ollama_client.py	Add sentence-transformers path for `ollama_embed` and `ollama_embed_batch` . Model loaded once at startup when flag is false.
requirements.txt	Add `sentence-transformers`

5. Expected Performance Improvement

Metric	Ollama (Current)	sentence-transformers
Embedding speed per chunk	200–500ms (HTTP overhead)	**<5ms (direct GPU)**
Batch support	Sequential (1 chunk per call)	**256 chunks per batch**
7,929 SMAC files	Multi-day (with timeouts)	**Under 1 hour**
Read timeouts	Frequent on large files	**None — no HTTP involved**
GPU utilization	Low (idle between HTTP calls)	**High (continuous batch processing)**

6. Step-by-Step Execution Plan

PHASE 1 — Local Laptop (Test First)

> **Step 1:** Install sentence-transformers
>
> ```
> pip install sentence-transformers
> ```

> **Step 2:** Download and cache the model (requires internet)

```
python -c "
from sentence_transformers import SentenceTransformer
model = SentenceTransformer('mixedbread-ai/mxbai-embed-large-v1')
vectors = model.encode(['test sentence'], normalize_embeddings=True)
print(vectors.shape)  # Should print (1, 1024)
"
```

Step 3: Code changes made to 3 files (config.py, ollama_client.py, requirements.txt)

Step 4: Test locally — set `USE_OLLAMA_EMBEDDINGS=false` in `.env` , start backend, upload one SMAC file, verify indexing and Q&A work

Step 5: Compare vectors — embed the same text with both methods, verify both return 1024 dimensions

PHASE 2 — Server Deployment (Requires Brief Internet Connection)

Internet required for Steps 6 & 7 only. After that, everything runs fully offline.

Step 6: Install sentence-transformers on server

```
source /opt/isd/venv/bin/activate
pip install sentence-transformers
```

Step 7: Pre-download the model (caches to `~/.cache/huggingface/`)

```
python -c "from sentence_transformers import SentenceTransformer; SentenceTransformer('mixedbread-ai/mxbai-embed-large-v1')"
```

Step 8: Copy updated code files to server (3 files)

- `config.py`
- `ollama_client.py`
- `requirements.txt`

Step 9: Update `.env` on server

```
USE_OLLAMA_EMBEDDINGS=false
```

Step 10: Stop current indexing

```
Ctrl+C
```

Step 11: Clear old data

```
rm -rf /opt/isd/ISDDocumentIntelligence_V5/backend/chroma_db_v5
rm /opt/isd/ISDDocumentIntelligence_V5/dbscripts/.smac_bulk_progress.db
mysql -u root -p -e "DELETE FROM ISDIntelligence.smac_reports;"
```

Step 12: Restart backend

```
source /opt/isd/venv/bin/activate
cd /opt/isd/ISDDocumentIntelligence_V5/backend
uvicorn app:app --host 0.0.0.0 --port 8001
```

Step 13: Re-index all SMAC files

```
source /opt/isd/venv/bin/activate
cd /opt/isd/ISDDocumentIntelligence_V5/dbscripts
python bulk_index_smac.py --folder "/data/SMAC Data" --case-id 0 --username rajibds --password rajibds --workers 5
```

7. Rollback Plan

If anything goes wrong, set `USE_OLLAMA_EMBEDDINGS=true` in `.env` and restart the backend. The system reverts to Ollama embeddings immediately. No code changes needed.

8. Key Points

Point	Detail
Internet needed	Only for Steps 6 & 7 (install package + download model). Fully offline after that.
Ollama still used	For LLM calls (gemma3:12b Q&A, entity extraction). Only embeddings change.
Toggle is reversible	Set `USE_OLLAMA_EMBEDDINGS=true` to revert to Ollama at any time.
Re-index required	Yes — vectors from different methods may differ slightly. Clean re-index ensures consistency.
Model is identical	Both methods use mxbai-embed-large-v1. Same architecture, same 1024 dimensions.
ChromaDB unchanged	Same vector store, same collections, same metadata. Only the embedding source changes.

> **Decision:** Test locally first (Phase 1), then deploy to server with brief internet window (Phase 2). Expected improvement: 7,929 files indexed in under 1 hour instead of multiple days. Fully reversible via .env toggle.

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